Subject: Mathematics | **CBSE CLASS 10 – WEEKLY TEST 05** | **Time:** 45 Minutes
Class: Class X (CBSE) | **Chapter 9: Some Applications of Trigonometry** | **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choice is provided in Question 10(iii).
2. **Section A** comprises 4 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 1 Very Short Answer question (2 marks) and 3 Short Answer questions (3 marks each).
4. **Section C** comprises 1 Applied Case Study question of 4 marks with internal choice.
5. Use of calculators is not permitted. Take $\sqrt{3} = 1.732$ unless stated otherwise.

SECTION A: Multiple Choice & Assertion-Reason (5 × 1 = 5 Marks)

Q1. From a point on the ground which is 30 m away from the foot of a tower, the angle of elevation of the top of the tower is 30°. The height of the tower is: [1]

(A) $10\sqrt{3}$ m (B) $20\sqrt{3}$ m (C) $30\sqrt{3}$ m (D) 10 m

Q2. Two vertical poles of heights 6 m and 11 m stand on flat ground. If the distance between their feet is 12 m, the distance between their tops is: [1]

(A) 11 m (B) 12 m (C) 13 m (D) 14 m

Q3. If the altitude of the sun changes from 30° to 60°, the length of the shadow of a 15 m high pole: [1]

(A) increases by $10\sqrt{3}$ m (B) decreases by $10\sqrt{3}$ m (C) decreases by $5\sqrt{3}$ m (D) remains same

Q4. If the height of a tower and the distance of the point of observation from its foot both are increased by 10%, then the angle of elevation of its top: [1]

(A) increases (B) decreases (C) remains unchanged (D) is halved

Q5. **Assertion (A):** If the height of a vertical tower is doubled, the angle of elevation of its top from a fixed point on the ground does not double.
Reason (R): The value of $\tan\theta = \frac{h}{d}$ doubles when h doubles, but $\tan(2\theta) \neq 2\tan\theta$ for acute angles. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

SECTION B: Short Answer Questions (11 Marks)

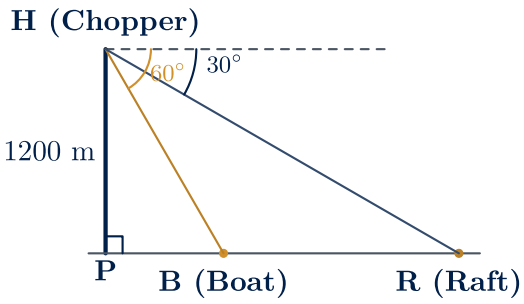
Q6. If a tower 30 m high casts a shadow $10\sqrt{3}$ m long on the ground, calculate the angle of elevation of the sun. [2]

Q7. A vertical flagstaff of height 5 m stands on top of a tower. From a point on the ground, the angles of elevation of the top and bottom of the flagstaff are 60° and 45° respectively. Find the height of the tower. [Take $\sqrt{3} = 1.732$] [3]

Q8. An observer on top of a cliff 100 m high observes two boats in the sea on opposite sides of the cliff. If the angles of depression of the boats are 30° and 45°, find the distance between the two boats. [3]

Q9. A straight highway leads to the foot of a tower. A man standing at the top of the tower observes a car at an angle of depression of 30°, which is approaching the foot of the tower with a uniform speed. Six minutes later, the angle of depression of the car is found to be 60°. Find the time taken by the car to reach the foot of the tower from this point. [3]

Q10. CASE STUDY: Coast Guard Airborne Search and Rescue Operation



During an emergency offshore maritime rescue mission, an Indian Coast Guard helicopter hovers stationary at an altitude of 1200 m directly above an island checkpoint (H). The flight tactical officer spots a stranded life raft (R) at an angle of depression of 30°. Simultaneously, a rescue speed-boat (B) is speeding towards the life raft along the same radial line. The angle of depression of the speed-boat from the helicopter is measured to be 60°. The point on the water directly beneath the helicopter is P.

Based on the above mathematical scenario, answer the following questions:

(i) Find the horizontal distance between the checkpoint P and the rescue speed-boat B. [1]

(ii) Find the direct line-of-sight distance from the helicopter H to the life raft R. [1]

(iii) Calculate the distance between the speed-boat and the life raft (i.e., length BR). If the boat travels at a speed of 20 m/s, how many seconds will it take to reach the raft? [Take $\sqrt{3} = 1.732$] [2]

— OR —

If the helicopter ascends vertically by 300 m to reach an altitude of 1500 m, what will be the horizontal distance to an object observed at an angle of depression of 45°? [2]

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